

Planetary Defense

Campaign Director • International NEO Response

9 PHASES • 40 STOPS • GRADE 6 • PLANETARY_DEFENSE_MS

THE OPENING CARD

Four hours ago a survey telescope found a faint dot moving against the stars. Four hours is not long enough to know where it is going, and some of the paths that fit go through the Earth. You run the campaign. The telescopes that watch it are yours, and so is the radar dish that gets one window in eleven years. Nobody knows yet whether this rock is the width of a street or of a town. That is the difference between one city and one country, and the people in both are waiting on what you can find out in time.

The Discovery Image

BEFORE THE PHASE — GREET

Two shifts that barely meet, and you are on both

The day crew work up what the night crew found, and the two hand over in eleven minutes at dawn. Put a name to most of them before your first handover. The survey runs on trusting a measurement somebody else took.

PLAN CARD 73W

Eleven days before the object passes Earth. Aino Virtanen's survey has flagged a faint dot that moved between three pictures. The same patch of sky holds forty thousand catalogued stars, two satellite streaks and a column of hot pixels. Today you decide whether there is really something out there, and say where to point next. Nothing else is known yet. Every hour of waiting widens the patch of sky it could be in.

BRIEFING 30W

A survey telescope flags a faint moving point near the ecliptic. The player must determine whether it is a real Solar System object, an image artifact, or a known source.

WHAT YOU DECIDE 11W

Validate a new-object candidate and preserve the measurements needed for follow-up.

1.1 Real object, or camera fault?

DISCOVERY & IMAGING · SURVEY TELESCOPE · A ROOM · CHOICE

SCENE 29W

Three survey pictures show the same faint dot in three different places. The background stars have not moved. A telescope on another mountain has just found the dot too.

GUIDE 57W

Four options, and the test is whether the thing moved in the sky or on the chip. Ask of each what it would leave behind. A cosmic ray arrives once. A hot pixel stays on its own pixel. Lining the pictures up wrongly would move the stars too. And another mountain has just found the same dot.

ASK 16W

The dot moved and the stars did not. What does that tell you about the dot?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a detector is the chip inside the camera, and its faults stay in one spot on the chip

VERDICT 74W

A real object moves against the sky. A camera fault moves with the camera. That is the test, and every reading here uses it. The dot is round in all three pictures, so it is not a cosmic ray, which hits once and hits sharp. It lands on a new pixel each time, so it is not a hot pixel. A second telescope found it, so it is not one camera making things up.

TAKEAWAY 16W

Something real moves against the stars, and a camera fault stays where the camera put it.

1.2 Check it before you send it

DISCOVERY & IMAGING · SURVEY TELESCOPE · A PERSON · SEQUENCE

SCENE 41W

Forty thousand stars, two satellite streaks and a bad column of pixels share the pictures with the moving dot. The camera's calibration is the check against known stars that says what its numbers mean. The discovery notice goes out at dawn.

GUIDE 55W

All four steps happen before dawn, so ask what each one rules out. The pictures are crowded with things that are not the dot: stars, streaks and a bad column. Ask of each step whether it separates the dot from the camera, or from something already known. And the last one spends somebody else's night.

ASK 12W

Put the four steps in the order they have to be done.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a prediction other people can test is what makes a claim checkable

VERDICT 68W

Each step throws out a different kind of mistake. Cleaning removes what the camera itself adds. Matching removes the satellites, the known faults and the objects already on the list. Measuring turns a picture into numbers, with a range that says how sure you are. The last step turns all of that into a prediction. A prediction can fail, and that is what makes the claim worth sending.

TAKEAWAY 12W

A claim is worth sending out once somebody else can test it.

1.3 How fast is it moving?

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A ROOM · BALLPARK

SCENE 29W

The dot shifted between two pictures taken twenty minutes apart. The next telescope has to be told where to point. Nobody has turned that shift into a rate yet.

GUIDE 54W

Five numbers, and two of them are the same gap written in different units. Another is the number of exposures. Ask of each whether this rate needs it. And note what the answer is for. It tells the next telescope where to point. It does not say how fast the rock is really going.

ASK 10W

About how fast is it moving, in arcseconds per hour?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a rate is an amount divided by the time it took

VERDICT 70W

Angular speed is a rate, so the units have to match before anything is divided. Six tenths of an arcsecond a minute is thirty-six arcseconds an hour, because an hour holds sixty minutes. This is not how fast the rock really flies. The same shift in the sky means very different true speeds at different distances. What it does tell you is how far the dot drifts between tonight's pictures.

TAKEAWAY 14W

Two pictures and a clock turn movement into a rate you can point with.

END OF THE PHASE 17W

Use digital images, signal-to-noise, artifacts, moving-object detection, controls as an evidence chain rather than as isolated facts.

Confirm the Motion

BEFORE THE PHASE — TRIAL

Walk the camp while it is still light

Everything here happens in the dark, and the camp is seven buildings on a ridge with no street lights. Walk it once in daylight, because a route learned in the light is one you can take at two in the morning with a red torch.

PLAN CARD 72W

Ten days to the pass. Six observatories have sent positions to Hugo Delacroix, who handles the campaign's measurements. Two used a different clock. One sent no pointing at all. One camera's positions bend away from the orbit. Today you build one clean track out of all of it, and buy the next observation. A weak track loses the object after the pass, and it will be too faint to find for years.

BRIEFING 24W

Follow-up telescopes obtain positions with different clocks and plate solutions. The player must combine astrometry without creating a false trajectory from inconsistent reference frames.

WHAT YOU DECIDE 14W

Produce a consistent astrometric track and identify observations that should be down-weighted or repeated.

2.1 Why do the positions disagree?

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · CHOICE

SCENE 29W

One wide camera's positions bend away from the orbit near one edge of its chip. The clock checks out. A second telescope, watching the same object, shows nothing odd.

GUIDE 55W

Four options, and the misses appear at one edge of one chip. Ask of each whether it would show up everywhere or only there. A rock speeding up would look the same from every telescope. Wrong times would too. A wrong star list would spoil every camera. What is left belongs to this camera alone.

ASK 17W

The misses appear at one edge of one camera's chip, and nowhere else. Where is the fault?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

the reference stars in a picture are a check on the picture itself, because they do not move

VERDICT 75W

The reference stars are the control. They do not move, so anything that shifts them belongs to the equipment. Here they miss by more toward one edge of the chip, and the asteroid misses the same way. That is a pattern in chip position, not in the sky. A real push would move the asteroid alone. The clock agrees, so it is not timing, and the second telescope agrees, so it is not the orbit.

TAKEAWAY 12W

An error that follows the equipment is telling you about the equipment.

2.2 Build the arc

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A ROOM · SEQUENCE

SCENE 24W

Six observatories have sent positions. Two used a different time standard, and one sent no pointing solution at all. Together they cover four hours.

GUIDE 51W

All four steps will happen. Six observatories have sent positions, two on a different clock and one with no pointing solution. Ask of each step what the next one would mean without it. And note what the goal is. Not more numbers, but numbers that would tell two possible paths apart.

ASK 12W

Put the four steps in the order they have to be done.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

an arc is the stretch of time your measurements cover

VERDICT 68W

A short arc fits many paths at once. They agree about where the object has been and disagree about where it goes. Time is what pulls them apart. Wait a few nights and paths that matched the same measurements spread out across the sky. So the real test is a prediction. Point where one path says the object will be, and the paths that were wrong are gone.

TAKEAWAY 19W

A path is tested by where the object turns up next, not by how well it fits the past.

2.3 Why waiting helps

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · CHOICE

SCENE 34W

Four hours of spare time on a two-metre telescope, and the object sets soon after dusk. Two possible paths fit tonight's measurements equally well. The same patch of sky comes back in six nights.

GUIDE 55W

Four options, and two paths fit tonight equally well. Ask of each what it would change about that. A sharper picture tonight measures the same moment more precisely. So does a bigger telescope, and so does taking more pictures in a row. What tells two paths apart is letting them drift away from each other.

ASK 13W

Two paths fit tonight equally well. What makes them easier to tell apart?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

two paths that agree tonight can disagree later

VERDICT 75W

Two paths through the same point tonight are heading slightly different ways. Tonight they sit on top of each other. No camera can split them, however sharp it is. Six nights later that small difference has become a real gap in the sky. One picture then does what ten pictures tonight cannot. So the booking worth having is the late one. A sharper look at tonight is worth much less than a rougher look later.

TAKEAWAY 21W

Two paths that agree today drift apart with time, so a later look separates them better than a sharper look now.

2.4 Read the risk correctly

ENTRY & CONSEQUENCES · ENTRY & CONSEQUENCES LAB · A ROOM · CLOUD

SCENE 30W

The best-guess path misses Earth, and the cloud of possible paths around it still touches the impact line. Dr. Marco Rossi, the orbit lead, wants one sentence before the briefing.

GUIDE 60W

The cloud on screen is every path the observations still allow. Its middle misses Earth. Its edge still touches the impact line, so this is not a stand-down yet. Two improvements are ready, and they do different things to the cloud: one moves it, one makes it smaller. Apply what you would authorise, then say what the briefing can state.

ASK 24W

What has to change before the whole cloud of paths clears Earth? Move the bars to show the middle and one sigma either side.

BEHIND THE BUTTON 160W

What the cloud is. The path is worked out from observations, and the observations allow a range of paths rather than one. Drawn on the encounter plane, that range is a cloud. The impact line is drawn on the same plane. Saying the best-guess path misses Earth is a statement about the middle of the cloud only.

Moving it and shrinking it are different jobs. Fixing an error in the best-guess path slides the whole cloud across without changing its size. The uncertainty is the same, just somewhere else. A new, well-placed observation adds information and makes the cloud smaller. Only that can clear Earth if it is the width that reaches the line.

Why the sentence matters. People hear safe or not safe. A stand-down that gets taken back costs the trust the next warning needs. So the briefing has to be about the whole cloud, and this is where you find out whether you can say that yet.

TAKES AS READ

an orbit fit gives a spread of possible paths, not one path · the middle of a spread can move without the spread getting any thinner

VERDICT 65W

An orbit is a spread of possible paths, not a dot. Fixing a known bias slides the whole spread sideways. It does not make the spread any thinner. Better measurements do make it thinner, but they do not choose where the middle sits. The call needs both. If a real part of the spread still crosses Earth, a safe-looking middle has not made anybody safe.

TAKEAWAY 12W

Risk lives in the whole spread, not in the middle of it.

2.5 Thirty-seven paths in a hundred thousand

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A PERSON · BALLPARK

SCENE 30W

The computer has finished flying every path that fits the measurements. What came back is a count. The number that leaves this room will be quoted in public for years.

GUIDE 51W

Four numbers, and two belong to other things. The first quick run, and the years to the encounter. Ask of each whether this share needs it. And note what the answer means. It is the share of the paths that still fit, and one new measurement can change it a lot.

ASK 9W

What is the chance of impact, as a percentage?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a chance can be worked out by counting how often something happens out of all the tries

VERDICT 71W

The computer draws many paths that all fit the measurements, then flies each one forward. Thirty-seven out of one hundred thousand land on Earth. That works out at 0.037 per cent. It is not a count of future impacts. It is the share of the spread that lands on us. So the number moves whenever the spread moves. One good measurement next week can shrink it to nothing, or double it.

TAKEAWAY 16W

A very small chance can only be measured by trying a very large number of times.

2.6 Point where the answers disagree

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · CHOICE

SCENE 34W

The best guess misses by four Earth radii, and about one path in six hundred still hits. Next Tuesday the two sets of paths sit eight arcseconds apart. After that they close up again.

GUIDE 52W

Four places to point, and a measurement only helps where the answers disagree. Ask of each option whether the hitting paths and the missing paths point somewhere different there. Before and after Tuesday they agree. Looking then confirms what nobody was arguing about. Brightness is not what decides this. A date is.

ASK 14W

A thin set of paths still hits Earth. Where do you point next week?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a measurement teaches you something where the possible answers disagree

VERDICT 65W

A measurement only teaches you something where the rival answers disagree. Here that is one night. On Tuesday the hitting paths and the missing paths sit eight arcseconds apart. Any follow-up telescope can tell those apart. Before and after, both sets predict the same spot. Point then and you confirm what nobody argued about. So the answer here is a date, not a better telescope.

TAKEAWAY 14W

The best measurement is the one whose result the rival answers would disagree about.

2.7 Moving with the sky, or not moving at all

DISCOVERY & IMAGING · SURVEY TELESCOPE · A ROOM · BELT

SCENE 31W

The survey found four hundred things last night and the report goes at dawn. Most of them are not objects. What tells them apart is how they move between the pictures.

GUIDE 38W

Two bins. A real object out there moves against the stars, the same way and the same amount in each picture. Everything else either does not move at all, or moves in a way nothing in space does.

ASK 14W

Send each detection to the bin that says whether something is really out there.

BEHIND THE BUTTON 84W

Why movement is the test. The stars stay put across one night. Anything in the solar system moves against them. Something in the same place in all four pictures is a star, a bad pixel or a reflection.

What the false ones look like. A cosmic ray shows up in one picture and never again. A satellite crosses so fast it draws a line. Reflections sit at a fixed place in the camera and move when the telescope moves, not when the sky does.

TAKES AS READ

the stars stay in the same places across one night

VERDICT 149W

A survey does not find objects. It finds differences between pictures, and the work is deciding which differences are things. A real object moves against the stars, the same direction and the same amount each time, which is why four pictures are taken rather than two. Three points can line up by chance and four almost never do. Everything else fails that test in its own way. A cosmic ray hits the camera once and is gone. A satellite crosses so fast it draws a line in a single picture. Hot pixels and reflections sit in fixed places in the camera, so they move when the telescope is nudged and stay put when the sky turns. None of this is about brightness, which is why a bright convincing blob is exactly the thing that wastes a follow-up night — and follow-up nights are what this survey has least of.

TAKEAWAY 15W

What makes a detection an object is how it moves, not how bright it is.

END OF THE PHASE 18W

Use angular coordinates, plate calibration, time standards, uncertainty, residuals as an evidence chain rather than as isolated facts.

How Large Is It?

BEFORE THE PHASE — FOLLOW

Fischer walks the cable run once

The dish's timing cable runs from the hut to the pad along a route nobody drew. Fischer, the radar dynamics lead, knows where it crosses the road. Stay with her, because where she stops is where the cable is buried shallow.

PLAN CARD 68W

Six days to the pass. Irina Sorokin has been waiting on telescope time all week. Everything anybody wants to know depends on the size, and the size rests on how bright the dot looked. Brightness alone allows anything from 120 metres to 600. At the small end this is a city. At the large end it is a country. Today you buy the measurement that breaks the tie.

BRIEFING 23W

The object's brightness is known, but size depends strongly on reflectivity. The player must combine brightness, albedo, and thermal evidence to bound diameter.

WHAT YOU DECIDE 10W

Produce a diameter range rather than a single unsupported value.

3.1 What makes it look bright?

PHYSICAL CHARACTERIZATION · SPECTROSCOPY DOME · A ROOM · PROTOCOL

SCENE 33W

The object was bright enough for the survey to spot it, and people across the room are reading that one number as a size. Four things that change brightness are on the board.

GUIDE 52W

Four changes, and four things they do to the brightness. Pair them by asking where in the journey each one acts. How much sunlight arrives? How much is sent back? Or how much of that reaches us? Brightness is all of them together, which is why one number cannot give a size.

ASK 12W

Match each change to what it does to the brightness we measure.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

light spreads out as it travels, so the same lamp looks fainter from further away

VERDICT 71W

Brightness is four things multiplied together, which is why size and surface get tangled. A bigger rock catches more sunlight. A better reflector sends back more of what lands on it. Distance from the Sun sets how much sunlight arrives at all. Distance from us sets how much of what leaves the rock gets here. The same brightness fits a small pale rock nearby or a big dark one far out.

TAKEAWAY 9W

A bright asteroid is not always a big one.

3.2 Same brightness, different size

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · BALLPARK

SCENE 29W

Two rocks look just as bright as each other. One reflects four times as well. The infrared team wants the sizes compared before anybody says a number out loud.

GUIDE 60W

Five numbers, and three of them are wrong ways to use the reflectivity. Its square. One divided by it. And the brightness itself. Ask of each whether it respects the two steps. Four times the reflectivity needs a quarter of the area. And width is the square root of area, so a quarter of the area is half the width.

ASK 11W

How wide is the shinier rock, compared with the other rock?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

albedo is how well a surface reflects the light that lands on it · the area of a circle grows as the square of its width

VERDICT 66W

Brightness pins down the area facing us multiplied by how well it reflects, and nothing else. Four times the reflectivity at the same brightness needs a quarter of the area. Width is the square root of area, so a quarter of the area is half the width. That is the whole trap. Every size taken from brightness has a guess about the surface hidden inside it.

TAKEAWAY 12W

Brightness alone cannot give a size unless somebody has assumed a reflectivity.

3.3 Settle the size

PHYSICAL CHARACTERIZATION · SPECTROSCOPY DOME · A PERSON · DEGENERACY

SCENE 29W

Reflected light alone puts the rock somewhere between one hundred and twenty and six hundred metres across. The heat team now has an infrared reading from the same night.

GUIDE 53W

Two controls slide the size and the surface brightness along a whole family of pairs that all match the reflected light. See how wide that family is first. Then bring in the heat reading, which depends on the same two things in a different way, and watch the family shrink to one pair.

ASK 14W

Which width and which surface fit the reflected light and the heat at once?

BEHIND THE BUTTON 135W

Why reflected light alone cannot settle it. What you measure is how much sunlight comes back. That depends on how big the rock is and how shiny it is, multiplied together. A small shiny rock and a big dark one send back the same light.

What the heat reading adds. Sunlight that is not reflected is absorbed and comes back out as heat. So the heat depends on the size and on how much is absorbed, which is a different mix of the same two unknowns. Put the two together and both are fixed.

Why the size is the number everybody wants. The energy of an impact goes as the size cubed. A rock five times wider carries more than a hundred times the energy, which is the difference between one region and a continent.

TAKES AS READ

a warm object glows in infrared light, which our eyes cannot see · reflected brightness depends on the size and the surface together

VERDICT 69W

Reflected light depends on size and surface together, so many pairs fit it equally well. That is a tie, not a noisy measurement, and measuring the same light more carefully will never break it. Heat depends on the two the other way round. A dark rock absorbs more sunlight, so it runs warmer and glows more. Lay the two answers over each other and they cross in one place.

TAKEAWAY 15W

A tie between two answers is broken by a measurement that depends on them differently.

3.4 Real object, or camera fault?

DISCOVERY & IMAGING · SURVEY TELESCOPE · A ROOM · CALLBACK · CHOICE

SCENE 29W

Three survey pictures show the same faint dot in three different places. The background stars have not moved. A telescope on another mountain has just found the dot too.

GUIDE 57W

Four options, and the test is whether the thing moved in the sky or on the chip. Ask of each what it would leave behind. A cosmic ray arrives once. A hot pixel stays on its own pixel. Lining the pictures up wrongly would move the stars too. And another mountain has just found the same dot.

ASK 16W

The dot moved and the stars did not. What does that tell you about the dot?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a detector is the chip inside the camera, and its faults stay in one spot on the chip

VERDICT 74W

A real object moves against the sky. A camera fault moves with the camera. That is the test, and every reading here uses it. The dot is round in all three pictures, so it is not a cosmic ray, which hits once and hits sharp. It lands on a new pixel each time, so it is not a hot pixel. A second telescope found it, so it is not one camera making things up.

TAKEAWAY 16W

Something real moves against the stars, and a camera fault stays where the camera put it.

END OF THE PHASE 18W

Use brightness, inverse-square law, albedo, thermal emission, size degeneracy as an evidence chain rather than as isolated facts.

What Is It Made Of?

PLAN CARD 67W

Five days out. A wide dip has appeared in the light from the rock. If it is real, it changes what Tola Adebayo has already written into a draft notice. Today you decide whether that dip belongs to the rock. This is not a label for a catalogue. What the rock is made of gives its weight, and its weight decides whether a spacecraft could move it.

BRIEFING 26W

Spectra show broad features, but weathering, grain size, and viewing geometry complicate composition. The player must avoid turning a taxonomic resemblance into an exact material claim.

WHAT YOU DECIDE 9W

Assign a composition class with explicit alternatives and confidence.

4.1 Whose dip is it?

PHYSICAL CHARACTERIZATION · SPECTROSCOPY DOME · A ROOM · TRACE

SCENE 42W

A wide dip has turned up in the spectrum, which is the rock's brightness measured colour by colour. It would change what the rock is made of. The room has a standard star, an air record, and a spectrum taken from space.

GUIDE 61W

Open each piece of evidence to see what the spectrum depended on. Keep what would still stand if the air were at fault, and untick the rest. Then name what the failing ones share. The dip would change what the rock is made of, so both halves count: keeping a false dip and throwing away a real one are equally wrong.

ASK 14W

Does the dip belong to the rock, to the air, or to the camera?

BEHIND THE BUTTON 123W

Why the air is the first suspect. Water vapour in Earth's air absorbs light at particular colours, and one of them is right where this dip is. Every ground telescope looks through that air, so every ground spectrum is the rock's spectrum multiplied by the air's.

What looking through more air tells you. If the dip gets deeper when the telescope looks through more atmosphere, the air is making it. If it stays the same, the air is not.

Why the space spectrum settles it. Above the atmosphere there is no water vapour in the way at all. It is the one measurement that does not go through the suspect step, which makes it worth more than any number of careful ground measurements.

TAKES AS READ

a spectrum is brightness measured colour by colour · air can leave marks of its own in a spectrum taken from the ground · airmass is how much air the light came through on the way in

VERDICT 66W

Three ground displays can agree because all three looked through the same air. The standard star has the dip too. The dip gets deeper when the light came through more air. The flat field is steady, so the camera is not adding it. And the spectrum taken from above the air has no dip at all. Agreement proves nothing when everything agrees for one shared reason.

TAKEAWAY 10W

Measurements that share one path do not check each other.

4.2 From light to what it is made of

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · SEQUENCE

SCENE 31W

The spectrum has two dips and a slope. Three different mineral mixes match it just as well as each other. A draft notice naming one of them is already being typed.

GUIDE 62W

All four steps will happen, and three mineral mixes fit just as well. Ask of each step what would be meaningless without it. The Sun, the air and the instrument all added something to what was recorded. A feature named before those come out could belong to any of them. And trying several mixes is what stops one being named too early.

ASK 12W

Put the four steps in the order they have to be done.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

what a telescope records includes everything the light passed through on the way

VERDICT 74W

Cleaning has to come first, because the Sun, the air and the instrument are all inside what the camera recorded. A dip found before they are removed may belong to any of them. Reflectance is then a property of the rock. Only after that is it worth asking which features hold up. Weathering and grain size can copy a change in minerals, so several models are tried and the report keeps what still fits.

TAKEAWAY 16W

A good answer about what something is made of keeps the other answers that still fit.

4.3 Make the claim stronger

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · CHOICE

SCENE 29W

The draft notice names what the rock is made of. One spectrum backs it. Next door the deflection team is sizing a spacecraft from the weight that class implies.

GUIDE 62W

Four options, and there is one spectrum, taken at one angle, at one point in the spin. Ask of each whether it repeats what has been done or brings something new.

Another spectrum the same way can repeat the same mistake. Next door somebody is sizing a spacecraft from this, so what is wanted is something that could rule a mix out.

ASK 14W

The rock turns as it goes. What does watching it at other angles add?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a measurement made one way can be wrong in the same way every time

VERDICT 69W

A firmer claim needs evidence that could disagree with the one you have. More colours reach features this spectrum cannot see at all, so whole families of minerals can be ruled out. Another look tonight, taken the same way, repeats the same mistakes. Checks that work by different physics come next, once the list is short. Keep the range visible, because the deflection team turns it into a weight.

TAKEAWAY 13W

Ask for evidence that could disagree with you, not for a firmer label.

4.4 Check it before you send it

DISCOVERY & IMAGING · SURVEY TELESCOPE · A ROOM · CALLBACK · SEQUENCE

SCENE 41W

Forty thousand stars, two satellite streaks and a bad column of pixels share the pictures with the moving dot. The camera's calibration is the check against known stars that says what its numbers mean. The discovery notice goes out at dawn.

GUIDE 55W

All four steps happen before dawn, so ask what each one rules out. The pictures are crowded with things that are not the dot: stars, streaks and a bad column. Ask of each step whether it separates the dot from the camera, or from something already known. And the last one spends somebody else's night.

ASK 12W

Put the four steps in the order they have to be done.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a prediction other people can test is what makes a claim checkable

VERDICT 68W

Each step throws out a different kind of mistake. Cleaning removes what the camera itself adds. Matching removes the satellites, the known faults and the objects already on the list. Measuring turns a picture into numbers, with a range that says how sure you are. The last step turns all of that into a prediction. A prediction can fail, and that is what makes the claim worth sending.

TAKEAWAY 12W

A claim is worth sending out once somebody else can test it.

END OF THE PHASE 17W

Use spectra, absorption features, mineralogy, taxonomy, model ambiguity as an evidence chain rather than as isolated facts.

Radar Contact

BEFORE THE PHASE — HUNT

Six lamps, and four are where the log says

Every telescope here is checked against portable lamps, and two went out with somebody last week. Until all six are found, nobody can say the brightness measurements are on one scale. Two scales invent objects that are not there.

PLAN CARD 69W

The pass is tonight, and Anna Fischer has the dish. The radar window is four hours wide, and this view does not come back for eleven years. Three of the four things the room wants can be got another way. Today you decide what radar alone can do, and write the order of the night. Tonight the size question closes for good, or it stays open for a decade.

BRIEFING 30W

A planetary radar window opens for several hours. Range and Doppler measurements could collapse the orbit and reveal shape, but only if the observation plan uses the limited signal effectively.

WHAT YOU DECIDE 11W

Design a radar sequence that improves both orbit and physical characterization.

5.1 What does radar measure?

RADAR & DYNAMICS · PLANETARY RADAR CONTROL · A ROOM · PROTOCOL

SCENE 32W

The radar room has its first clean run of echoes. Delay, pitch shift and echo width are up on one screen. The orbit team is waiting to hear which reading to use.

GUIDE 51W

Four things we want to know, and four radar readings. Pair them by asking what each reading really is. A time? A spread of pitch? A pattern of when the echo comes back? Or a set of pictures? One of them only exists after you have taken the others many times.

ASK 14W

Match each thing we want to know to the radar reading that measures it.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

radar sends a radio pulse out and listens for what comes back · a shift in the pitch of an echo tells you how fast something moves toward or away

VERDICT 70W

Radar pulls apart what a picture leaves tangled, and each reading pulls apart something different. The delay of the echo is a time, and with a known speed a time is a distance. The spread in pitch is the range of speeds across the rock, which is spin. Echo strength against time says how much surface sits at each depth. Only pictures taken right through a turn give the shape.

TAKEAWAY 14W

Each radar reading measures one thing, and none of them measures all of it.

5.2 How far away, from the echo?

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · BALLPARK

SCENE 24W

The dish sent a pulse, and the echo came back four seconds later. The characterisation team needs a distance before the next run starts.

GUIDE 51W

Five numbers, and two belong to other questions. The years to the encounter, and that night's seeing. Ask of each whether this distance needs it. And note the two in the sum. The pulse went out and came back, so the time covers the trip twice. Forgetting that doubles the answer.

ASK 7W

About how far away is the object?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

radio waves travel at a fixed speed, three hundred thousand kilometres every second

VERDICT 72W

Radar hands you a time, and a time becomes a distance as soon as the speed is known. Radio travels three hundred thousand kilometres each second. Four seconds is one million two hundred thousand kilometres of travelling. The pulse went out and came back, so the rock sits half that away. That is six hundred thousand kilometres. Radar is worth so much here because it measures a distance instead of fitting one.

TAKEAWAY 13W

An echo becomes a distance once you know the speed and the time.

5.3 What this pass does to the next one

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A ROOM · CHOICE

SCENE 26W

Radar has pinned this pass at 2.4 Earth radii and the room starts to relax. Then the return solution flashes up on the board: eleven years.

GUIDE 55W

Four options, and radar has settled this pass. Ask of each whether it says anything about the one in eleven years. Passing that close lets Earth's pull change the rock's path, so the orbit that comes back is not the one that arrived. Fresh measurements afterwards check that working out. They do not replace it.

ASK 17W

Radar has settled this pass. What has to be worked out before anybody calls the object safe?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a planet's gravity bends the path of anything that passes close to it

VERDICT 76W

A measured miss settles this pass and not the next one. Passing a few Earth radii out lets Earth's gravity pull on the rock and change its speed and its direction. That changes how long its trip round the Sun takes, so it returns at a different time and in a different place. A narrow band of today's paths can turn into a much closer pass later. Safety has to be worked out after the bend.

TAKEAWAY 19W

A close pass rewrites the path it happens on, so safety is a claim about the passes after it.

5.4 On target, while the wind gets up

RADAR & DYNAMICS · PLANETARY RADAR CONTROL · A PERSON · HOLD

SCENE 29W

The object is faint and the pass lasts eleven minutes. The dish has to stay pointed at it while wind comes down the ridge and pushes the dish about.

GUIDE 47W

Hold the pointing error inside the band on the console. The band narrows through the pass, because the object gets closer and the beam needs to be more exactly on it. Each gust keeps pushing for as long as it lasts, so hold the drive against it.

ASK 9W

Hold the dish on the object through the pass.

BEHIND THE BUTTON 65W

Why pointing decides everything. The signal is strongest in the middle of the beam and falls away quickly. A dish half a beam off is not getting a slightly worse measurement. It is getting noise.

Why a gust is not one push. Wind loads the dish for as long as it blows. The drive has to hold against it, not correct once and let go.

TAKES AS READ

a dish has to be pointed at something to receive from it

VERDICT 120W

The signal falls away from the middle of the beam, so pointing is not a detail. It is the difference between a measurement and eleven minutes of nothing. Every push here lasts. A gust off the ridge loads the dish while it blows, and a dish left alone is pushed off and stays off, so the drive has to hold against it rather than correct and let go. The band narrows because the object is getting closer through the pass and the measurement gets fussier about where the dish is looking. And there is no second try tonight. The pass is the pass. The next one is when the object comes round again, which for this one is years away.

TAKEAWAY 12W

A push that lasts is answered by holding, not by a nudge.

5.5 Read the risk correctly

ENTRY & CONSEQUENCES · ENTRY & CONSEQUENCES LAB · A ROOM · CALLBACK · CLOUD

SCENE 30W

The best-guess path misses Earth, and the cloud of possible paths around it still touches the impact line. Dr. Marco Rossi, the orbit lead, wants one sentence before the briefing.

GUIDE 60W

The cloud on screen is every path the observations still allow. Its middle misses Earth. Its edge still touches the impact line, so this is not a stand-down yet. Two improvements are ready, and they do different things to the cloud: one moves it, one makes it smaller. Apply what you would authorise, then say what the briefing can state.

ASK 24W

What has to change before the whole cloud of paths clears Earth? Move the bars to show the middle and one sigma either side.

BEHIND THE BUTTON 160W

What the cloud is. The path is worked out from observations, and the observations allow a range of paths rather than one. Drawn on the encounter plane, that range is a cloud. The impact line is drawn on the same plane. Saying the best-guess path misses Earth is a statement about the middle of the cloud only.

Moving it and shrinking it are different jobs. Fixing an error in the best-guess path slides the whole cloud across without changing its size. The uncertainty is the same, just somewhere else. A new, well-placed observation adds information and makes the cloud smaller. Only that can clear Earth if it is the width that reaches the line.

Why the sentence matters. People hear safe or not safe. A stand-down that gets taken back costs the trust the next warning needs. So the briefing has to be about the whole cloud, and this is where you find out whether you can say that yet.

TAKES AS READ

an orbit fit gives a spread of possible paths, not one path · the middle of a spread can move without the spread getting any thinner

VERDICT 65W

An orbit is a spread of possible paths, not a dot. Fixing a known bias slides the whole spread sideways. It does not make the spread any thinner. Better measurements do make it thinner, but they do not choose where the middle sits. The call needs both. If a real part of the spread still crosses Earth, a safe-looking middle has not made anybody safe.

TAKEAWAY 12W

Risk lives in the whole spread, not in the middle of it.

Use radar ranging, Doppler shift, resolution, signal-to-noise, inverse problems as an evidence chain rather than as isolated facts.

Impact Energy

BEFORE THE PHASE — CANVASS

Who saw the bright thing on Tuesday

Half the valley reported something bright on Tuesday night. The office has to say whether it was the re-entry everybody assumes. Ask around until enough accounts have come back to separate one bright thing from two.

PLAN CARD 70W

The object passed a week ago at four times the distance of the Moon, and it is heading away. What it left behind is a description: about 180 metres across, and roughly four hundred megatons if the return in eight years arrives. Rohan Banerjee wants one number for civil defence today. Adebayo says the width is still the widest guess. Today you give a low, a middle and a high.

BRIEFING 31W

The latest size and speed estimates imply regional consequences, but uncertainty in diameter, density, and velocity spans a wide range. The player must compute scales without presenting a false point estimate.

WHAT YOU DECIDE 10W

Provide low, central, and high impact-energy scenarios with transparent assumptions.

6.1 How much energy does it carry?

ENTRY & CONSEQUENCES · ENTRY & CONSEQUENCES LAB · A ROOM · BALLPARK

SCENE 30W

Civil defence will not call anyone in without an energy figure. The impact group has a weight for the rock. It also has the speed the rock would arrive at.

GUIDE 54W

Four numbers, and two of them belong to other questions. How wide it is, and how many joules make a megaton. Ask of each whether this energy needs it. The weight and the speed are what the sum uses. And the speed goes in twice over, which is why speed matters more than weight.

ASK 8W

About how much energy would the impact carry?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the energy of a moving thing depends on how heavy it is and how fast it goes

VERDICT 71W

The energy of a moving thing is half its weight, multiplied by the speed twice over. That second speed is what makes this number so touchy. Double the speed and the energy is four times bigger. Double the weight and it only doubles. So speed and size are not equally worth measuring. It is also why a rock the size of a house does what nothing that size does down here.

TAKEAWAY 14W

Speed counts twice over in energy, so it is the number worth measuring best.

6.2 Twice as fast

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · CHOICE

SCENE 32W

The arrival speed in the briefing has just doubled after a new tracking run. The energy in that briefing came from the old speed, and it is read out in an hour.

GUIDE 56W

Four options, and only the speed has changed. Ask of each how many times the speed enters the energy. Twice the speed gives twice the momentum and more than twice the energy. One option talks about the air, and one about the crater. Neither of those is the question. The briefing goes out in an hour.

ASK 15W

The rock arrives twice as fast as the briefing assumed. What happens to the energy?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

the energy of a moving thing uses its speed twice over

VERDICT 70W

The energy of a moving thing is half the weight, times the speed twice over. Weight goes in once, so doubling it doubles the energy. Speed goes in twice, so doubling it multiplies the energy by four. A briefing built on the old speed is not slightly wrong. It is wrong four times over, and the warning area, the number of people and the whole plan were sized from it.

TAKEAWAY 18W

A quantity that appears twice in a rule moves the answer much further than one that appears once.

6.3 Which measurement is worth the most?

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · CHOICE

SCENE 29W

The range of harm runs from one region to a whole continent. Every part of that range comes from a measurement. There is time to improve one of them.

GUIDE 54W

Four options, and every part of that range came from a measurement. Ask of each how strongly it feeds the energy. Weight grows with width three times over, so a small error in width is a big error in weight. The speed goes in twice. And one option refuses to narrow anything at all.

ASK 11W

The range of consequences is enormous. Which measurement narrows it most?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a number that is multiplied in three times spreads any error three times as wide

VERDICT 70W

Weight comes from volume, and volume is the width multiplied in three times over. So an error in the width arrives at the energy three times as wide as it started.

Density goes in once. Speed goes in twice and is already the best known of the three, so there is least left to win there. That is why the width is worth more than anything else on this list.

TAKEAWAY 19W

Improve the measurement that carries the most error into the answer, not the one that is easiest to repeat.

6.4 How fast is it moving?

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A ROOM · CALLBACK · BALLPARK

SCENE 29W

The dot shifted between two pictures taken twenty minutes apart. The next telescope has to be told where to point. Nobody has turned that shift into a rate yet.

GUIDE 54W

Five numbers, and two of them are the same gap written in different units. Another is the number of exposures. Ask of each whether this rate needs it. And note what the answer is for. It tells the next telescope where to point. It does not say how fast the rock is really going.

ASK 10W

About how fast is it moving, in arcseconds per hour?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a rate is an amount divided by the time it took

VERDICT 70W

Angular speed is a rate, so the units have to match before anything is divided. Six tenths of an arcsecond a minute is thirty-six arcseconds an hour, because an hour holds sixty minutes. This is not how fast the rock really flies. The same shift in the sky means very different true speeds at different distances. What it does tell you is how far the dot drifts between tonight's pictures.

TAKEAWAY 14W

Two pictures and a clock turn movement into a rate you can point with.

END OF THE PHASE 23W

Use mass and density, kinetic energy, scaling, how uncertainty carries forward, orders of magnitude as an evidence chain rather than as isolated facts.

Can We Move It?

PLAN CARD 77W

Four months after the pass. Anna Fischer has the best orbit this campaign will ever have, and it is already two months old. The return is seven and a half years out. That sounds like plenty, and it is not, because a deflection has to launch years before it arrives. Today you size the push and decide how success gets measured. On a weak rock, a hard push breaks it into several pieces on the same path.

BRIEFING 30W

Impact probability remains significant, and a scouting mission could arrive years before encounter. The player must compare momentum transfer, lead time, and uncertainty without treating deflection as a movie-style explosion.

WHAT YOU DECIDE 11W

Choose a deflection demonstration and define how success will be measured.

7.1 How much does the push move it?

ENTRY & CONSEQUENCES · ENTRY & CONSEQUENCES LAB · A ROOM · BALLPARK

SCENE 29W

The deflection lab is sizing a spacecraft. It knows the craft's weight and how fast it would close, and the rock's weight to within a factor of a few.

GUIDE 53W

Four numbers, and two of them belong to an earlier step. The millimetres a second, and the extra push from thrown-off material, were used to get the yearly figure. Ask which two this sum needs. And note why a tiny nudge works at all. It gets multiplied by every year that is left.

ASK 14W

How far will the nudge have moved the rock by the time it arrives?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

momentum is weight multiplied by speed, and a collision passes it on · the same push shared over more weight makes less change in speed

VERDICT 94W

The push is tiny. A craft weighing ten tonnes hits a rock weighing eight billion, so the rock's speed changes by about fifteen millimetres a second — walking pace divided by two hundred. What makes it matter is that the rock keeps that speed, second after second, for eight years. Fifteen millimetres a second works out at about 475 kilometres in a year, and eight of those is thousands of kilometres of sideways shift. That is the whole idea of a nudge: it is not the push that is big, it is the time.

TAKEAWAY 11W

The same push moves a heavier thing less, in exact proportion.

7.2 What makes a deflection work?

ENTRY & CONSEQUENCES · ENTRY & CONSEQUENCES LAB · A PERSON · PROTOCOL

SCENE 28W

The impactor plan is up on the mission board. Beside it sit the weight estimate and the tracking plan. The director wants it all sorted before the review.

GUIDE 56W

Four items on the board, and four effects. Pair them by asking which part each one changes. The push that lands? How much that push moves the rock? Or how far the rock has gone by the time it arrives? One of the four is not a prediction at all. It is a measurement, taken afterwards.

ASK 13W

Match each item on the board to what it does to the outcome.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

a tiny change in speed adds up to a large change in position over years

VERDICT 68W

A deflection has four separate parts, and only one of them is the collision. The push that lands depends on the craft's weight and closing speed. What that push does depends on the rock's weight. Material thrown off the surface adds more push, by an amount nobody can predict yet. Time turns a tiny speed change into distance. Without tracking afterwards none of it is measured, only hoped.

TAKEAWAY 13W

A deflection is an experiment, and the thing it changes is the orbit.

7.3 Say what you expect, then measure it

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A ROOM · VERIFY

SCENE 24W

The impactor has hit and the control room is cheering. Nobody has measured the orbit yet. One tracking campaign is left in the budget.

GUIDE 46W

Write down the change you expected before the impact, and lock it. Then run the tracking campaign and read what the orbit actually did. The cheering is not the result. One campaign is left in the budget, so this is the only chance to find out.

ASK 14W

Did the measured change in the orbit match what was predicted before the impact?

BEHIND THE BUTTON 133W

Why hitting it is not the same as moving it. The spacecraft changes the rock's speed by a few millimetres a second. How much depends mostly on the rubble thrown off the surface when it hits, and nobody knows what that surface is made of. So the size of the push has to be measured, not assumed.

What the campaign measures. Where the rock was going before, and where it is going now, watched long enough that the difference is bigger than the measurement error. That is what turns 'we hit it' into 'we moved it this much'.

Why the prediction is written first. If the number is written after the measurement, it becomes whatever makes the mission look right. Then the one thing this whole mission existed to find out is lost.

TAKES AS READ

the momentum sum gives the smallest change in speed to expect · material thrown off the surface can add to the push

VERDICT 65W

The momentum sum gives a smallest change of about one and a half centimetres a second. Material thrown off the surface can add to that. So the prediction has to be a range, not one number. A hit proves the craft arrived. It does not measure the new orbit. Only tracking does that. Comparing it against a locked prediction tests the physics, not the aim.

TAKEAWAY 11W

An action is not finished until somebody measures what it did.

7.4 Why do the positions disagree?

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · CALLBACK · CHOICE

SCENE 29W

One wide camera's positions bend away from the orbit near one edge of its chip. The clock checks out. A second telescope, watching the same object, shows nothing odd.

GUIDE 55W

Four options, and the misses appear at one edge of one chip. Ask of each whether it would show up everywhere or only there. A rock speeding up would look the same from every telescope. Wrong times would too. A wrong star list would spoil every camera. What is left belongs to this camera alone.

ASK 17W

The misses appear at one edge of one camera's chip, and nowhere else. Where is the fault?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

the reference stars in a picture are a check on the picture itself, because they do not move

VERDICT 75W

The reference stars are the control. They do not move, so anything that shifts them belongs to the equipment. Here they miss by more toward one edge of the chip, and the asteroid misses the same way. That is a pattern in chip position, not in the sky. A real push would move the asteroid alone. The clock agrees, so it is not timing, and the second telescope agrees, so it is not the orbit.

TAKEAWAY 12W

An error that follows the equipment is telling you about the equipment.

END OF THE PHASE 17W

Use momentum, impulse, lead time, kinetic impact, uncertainty as an evidence chain rather than as isolated facts.

Evacuate or Wait?

BEFORE THE PHASE — EVADE

The visitor who wants a look through the telescope

A guest has been in the dome since dusk wanting to be shown something. Every minute the telescope is off target is sky nobody covers tonight. Keep out of his way and get your round done.

PLAN CARD 66W

Three years in. The impactor launched two years ago and reaches the asteroid in eleven days. The chance has moved twice this month, both times downward. Ellery's draft evacuation plan is already with two regional governments. Today you match what gets done to what is actually known. Stockpiles are cheap and can be undone. Moving nine million people is neither, and it kills people by itself.

BRIEFING 28W

The impact corridor crosses populated regions, but the probability is still changing. The player must connect scientific uncertainty to staged civil decisions and avoid both complacency and panic.

WHAT YOU DECIDE 11W

Create trigger-based preparations proportional to probability, consequence, lead time, and reversibility.

8.1 Match what you do to what you know

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · PROTOCOL

SCENE 27W

The regional team is round one table. The wall shows the newest corridor, the likely damage area and the time left. Today they choose what to do.

GUIDE 60W

Four states of the evidence, and four things to do. Pair them by asking three questions. How bad would it be? How long is left? And can the action be undone? Cheap, undoable things suit a long wait. Ordering people to move suits having run out of time. And saying so in public is what makes the next warning believed.

ASK 11W

Match each state of the evidence to what it justifies doing.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

a reversible action can be undone afterwards; an irreversible one cannot

VERDICT 78W

What you do depends on three things: how bad it would be, how long you have, and whether you could undo it. Years in hand means there is still time to learn more, so watch and prepare cheaply. A rising chance is worth real preparation, as long as it can be undone. With days left, the time to learn more has gone. And a chance that falls is stood down out loud, or the next warning is ignored.

TAKEAWAY 11W

You can prepare in stages without ever claiming to be certain.

8.2 How many people is that share?

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · BALLPARK

SCENE 31W

The emergency office needs a number to plan against. It has the people living inside the corridor. It has today's chance of impact. And it has a form to fill in.

GUIDE 58W

Five numbers, and three belong to other sums. The biggest city, the earlier chance, and the paths that hit. Ask of each whether this one needs it. A share of a whole means multiplying, once the share is written as a decimal. What comes out is a planning figure, not a number of people who will really move.

ASK 8W

How many people does that share come to?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a percentage is a share out of one hundred

VERDICT 69W

This number is a share, not a headcount. Half of one per cent of nine million people is forty-five thousand. Nobody expects exactly that many people to move. Either the rock misses and nobody moves, or it hits and far more than that do. What the share is good for is comparing plans. It puts the cost of preparing beside the risk of not preparing, in the same units.

TAKEAWAY 17W

A small share of a very large number is still a number somebody has to plan for.

8.3 Prepare without panic

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · TRIGGER

SCENE 29W

Five orbit updates are queued. Planners can still prepare without moving anybody. Both planning and supplies take time, and the thresholds have to be set before the stream starts.

GUIDE 75W

One rule, written before the next updates arrive. Once the board is released the number cannot change. Starting the planning cell takes two days from the moment it is called, and the last update arrives with no time left at all. So your line has to be crossed early enough for those two days to fit — and not so early that a cell is started for an object that was never going to matter.

ASK 17W

At what impact probability do you start up the regional planning cell, when it takes two days?

BEHIND THE BUTTON 127W

Why set the rules first. The updates arrive one after another, each one changing the odds a little. A threshold changed while they come in is just a description of what somebody already decided, and it cannot be explained afterwards to the people who acted on it.

Why two numbers rather than one. Writing a plan takes weeks. Moving supplies takes days, and needs the plan to exist first. One threshold set for the slow job fires far too early for the fast one, and set for the fast one leaves no time to plan.

What preparing without panic buys. Preparation can be undone cheaply. Evacuating people cannot. Good thresholds keep the cheap actions available long enough that the expensive one never becomes the only one left.

TAKES AS READ

the chance of impact moves as new measurements arrive · preparations take time to set up, and some of them can be undone

VERDICT 70W

A threshold is a decision made before the next number lands. Preparation takes time, so an action that is right can still be useless if it starts too late. Setting the rule first also stops every new number being judged by a new standard. Early thresholds go on things that can be undone. Later ones go on the things that cost more. None of it pretends the chance is certain.

TAKEAWAY 15W

A rule written before the numbers arrive is the only rule the numbers cannot bend.

8.4 What the telescope is chasing tonight

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · SPOT

SCENE 24W

The target list runs all night. One priority says which to observe, and whoever has the desk changes it as the night goes on.

GUIDE 43W

Observe what the priority asks for and leave the rest on the list. The board changes during the night and nobody announces it. What counts is the targets either side of a change, because a target that has set cannot be brought back.

ASK 12W

Observe to the priority on the board, and keep checking the board.

BEHIND THE BUTTON 83W

Why the priority changes. A new object nobody has tracked yet is the most urgent thing on any night, until enough measurements exist to predict where it goes. Then it becomes whatever is setting soonest, because a target below the horizon cannot be observed at all.

Why a missed target is different from other mistakes. Most work can be done tomorrow. An object low in the sky is up for an hour a night for two weeks, and then not for a year.

TAKES AS READ

an object can only be observed when it is above the horizon

VERDICT 114W

Three priorities run across the night and each wants different targets. Anything nobody has tracked yet, then anything setting soon, then anything with a deadline. A target can match two at once, and that is what makes a change cost something.

What the panel scores is the targets either side of a change. Most of the list is wanted by neither priority, so leaving those alone proves nothing. This is the case where the cost is highest, because the sky does not wait. An object low in the west is up for an hour a night for a fortnight, and a telescope spent on the last hour's priority adds a year to somebody's orbit.

TAKEAWAY 14W

A rule that has changed costs most when the chance does not come back.

8.5 Build the arc

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A ROOM · CALLBACK · SEQUENCE

SCENE 24W

Six observatories have sent positions. Two used a different time standard, and one sent no pointing solution at all. Together they cover four hours.

GUIDE 51W

All four steps will happen. Six observatories have sent positions, two on a different clock and one with no pointing solution. Ask of each step what the next one would mean without it. And note what the goal is. Not more numbers, but numbers that would tell two possible paths apart.

ASK 12W

Put the four steps in the order they have to be done.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it.

Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

an arc is the stretch of time your measurements cover

VERDICT 68W

A short arc fits many paths at once. They agree about where the object has been and disagree about where it goes. Time is what pulls them apart. Wait a few nights and paths that matched the same measurements spread out across the sky. So the real test is a prediction. Point where one path says the object will be, and the paths that were wrong are gone.

TAKEAWAY 19W

A path is tested by where the object turns up next, not by how well it fits the past.

END OF THE PHASE 19W

Use risk thresholds, expected consequence, decision under uncertainty, communication, reversibility as an evidence chain rather than as isolated facts.

The Final Defense Review

BEFORE THE PHASE — TAG

Catch Rossi before the fit starts

Rossi, the orbit determination lead, has two extra measurements from Monday that never went into the file. Once the orbit fit is running it takes two hours and he will not stop it. Two points change where this object is predicted to be.

PLAN CARD 71W

Eight years after the discovery, and the encounter is next week. The impactor struck five years ago, and tracking since then puts the object about two Earth radii clear. Some five thousand kilometres of that was bought by the impact. Nothing further can be done now, which is as it should be. Today you take the claims one at a time with Isabel Garcia, and write the public record with Ellery.

BRIEFING 21W

The international board must decide whether the object is sufficiently characterized, whether the deflection worked, and which civil preparations remain necessary.

WHAT YOU DECIDE 14W

Produce a claim-by-claim planetary-defense decision package and preserve monitoring after the immediate threat passes.

9.1 What each claim has earned

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · PROTOCOL

SCENE 31W

The board has to publish one account of an eight-year campaign. The four claims in front of it rest on very different evidence. Whoever runs the next one will read it.

GUIDE 52W

Four claims and four things to do about them. Pair them by asking what state each claim is in. Settled? Still open? Out of date? Or bigger than this rock? Giving four different states the same confidence is how a report stops being trusted. Whoever runs the next campaign reads this one.

ASK 11W

Match each claim to what the evidence under it justifies doing.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

a conclusion is only as strong as the evidence under it

VERDICT 73W

These four claims are in four different states, and giving them one level of confidence is how a report stops being trusted. A shift that somebody else has confirmed can be accepted. An edge nobody has pinned down is a reason to keep watching, not a reason to quote the middle. Plans built on an old corridor have to be fixed before use. And the gaps in the search outlive this rock entirely.

TAKEAWAY 11W

Different claims in one document have earned different amounts of confidence.

9.2 Signed is not checked

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A ROOM · ATTEST

SCENE 28W

Two independent checks are left before the record goes public tonight. Every claim in it is signed. Some of them would change the conclusion if they were wrong.

GUIDE 52W

Two checks left, and the record goes out tonight. Open each signed claim and read what actually backs it. Hold the ones the evidence does not support, and spend the checks where a wrong claim would do the most damage — because what goes out tonight is where the next team starts.

ASK 12W

Which signed claims have to be checked before the record goes out?

BEHIND THE BUTTON 104W

Why signed is not checked. A signature says somebody took responsibility for a sentence. It does not say anybody went and looked. Under a deadline, a guess and an observation end up written the same way on the same page.

Why a public record is harder to fix than to write. People will read it who cannot see the evidence, and quote it who were not there. A correction afterwards never travels as far as the original did.

How to pick two out of several. Not the shakiest. The one that would send the next team in the wrong direction if it were wrong.

TAKES AS READ

a signature records that somebody approved something · checking costs time, and there is not enough of it for everything

VERDICT 69W

A record is trustworthy when somebody else can rebuild what was measured and how. A signature says a person approved it. It does not say the thing is true. So the checks go to claims that matter and that nobody has confirmed another way. Checking an orbit fit that has already been reproduced spends a check and strengthens nothing. The weakest important link is the one worth the money.

TAKEAWAY 11W

A signature records a decision. It does not measure a thing.

9.3 What survives the success

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · CHOICE

SCENE 32W

The threat is gone, and so is most of the budget. The search still misses parts of the sky, and two follow-up sites lose their funding at the end of the year.

GUIDE 53W

Four options, and the threat is gone along with most of the money. Ask of each what it does to warning time. Finding a rock earlier buys nights for the path, time for the size, and years for a small nudge to work. Nothing later can give back time that was never bought.

ASK 10W

The campaign budget ends with this review. What survives it?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

warning time comes from finding things early, not from planning for them

VERDICT 71W

Warning time is what every later choice gets bought with. Finding a rock earlier gives more nights to check the orbit. It gives time to measure the size and what it is made of. It makes a small push worth giving, because years turn a push into distance. It lets people prepare in stages. Plans and models matter too, and none of them can give back time that was never bought.

TAKEAWAY 14W

The lasting defence is a system that can find, learn, decide and act again.

9.4 Moving with the sky, or not moving at all

DISCOVERY & IMAGING · SURVEY TELESCOPE · A ROOM · CALLBACK · BELT

SCENE 31W

The survey found four hundred things last night and the report goes at dawn. Most of them are not objects. What tells them apart is how they move between the pictures.

GUIDE 38W

Two bins. A real object out there moves against the stars, the same way and the same amount in each picture. Everything else either does not move at all, or moves in a way nothing in space does.

ASK 14W

Send each detection to the bin that says whether something is really out there.

BEHIND THE BUTTON 84W

Why movement is the test. The stars stay put across one night. Anything in the solar system moves against them. Something in the same place in all four pictures is a star, a bad pixel or a reflection.

What the false ones look like. A cosmic ray shows up in one picture and never again. A satellite crosses so fast it draws a line. Reflections sit at a fixed place in the camera and move when the telescope moves, not when the sky does.

TAKES AS READ

the stars stay in the same places across one night

VERDICT 149W

A survey does not find objects. It finds differences between pictures, and the work is deciding which differences are things. A real object moves against the stars, the same direction and the same amount each time, which is why four pictures are taken rather than two. Three points can line up by chance and four almost never do. Everything else fails that test in its own way. A cosmic ray hits the camera once and is gone. A satellite crosses so fast it draws a line in a single picture. Hot pixels and reflections sit in fixed places in the camera, so they move when the telescope is nudged and stay put when the sky turns. None of this is about brightness, which is why a bright convincing blob is exactly the thing that wastes a follow-up night — and follow-up nights are what this survey has least of.

TAKEAWAY 15W

What makes a detection an object is how it moves, not how bright it is.

END OF THE PHASE 19W

Use evidence synthesis, readiness, residual risk, international coordination, long-term stewardship as an evidence chain rather than as isolated facts.

THE CLOSING CARD

The object passed eleven days ago, two Earth radii out and falling further behind every hour. Most of that margin was bought by an impactor that hit five years earlier, and the radar window is what proved it had worked. Nobody evacuated Valle Seco. The school there reopened on the Monday.

The survey that found it is still running tonight, and the file stays open — this one is settled, and the next one has not been found yet.

Card text only — no options, boards or answers. 7,539 words of card prose across 40 stops. Generated from themes/planetary_defense_ms by tools/make-cardbook.mjs.